

* Machine Learning *

Date 4-18
Page

1. ~~The~~ learn How Machine Learning works :

The learning process

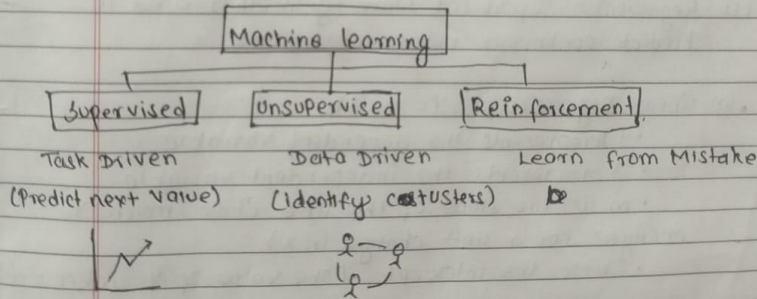
- (1) collect data : Gather examples (emails, img, pic)
- (2) Train Model : show data to algorithm
- (3) Learn patterns : Model finds relationships
- (4) Make predictions : use Model on new data

collect data → Train Model → Learn patterns → Predict

100 emails + labels	feed to algorithm	find spam Patterns	New mail Spam!
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→ machine learning means finding patterns in data and making prediction

Types of Machine learning



(i) Supervised Learning:

Supervised learning is a branch of Artificial intelligence that focus on training models to make predictions or decision based on labeled training data.

Definition:

Learn from labeled data (input $\rightarrow$ correct output)

How it works:

- You show computer: "This is a cat" (img + label)
- Computer learns: cats have whiskers, pointy ears, etc
- New img $\rightarrow$ computer predicts: "Cat!"

* Two Main Types:

Classification

- Predict Category
- Spam or Not Spam
- Cat, Dog, or Bird
- Sick or Healthy

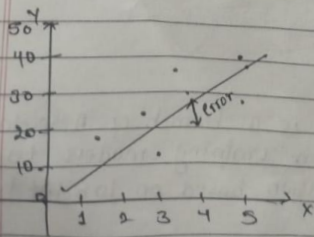
Regression

- Predict number
- House Price: \$350k
- Temperature: 25°C
- Age: 42 Yrs

(1) Regression: Supervised learning method used to Predict continuous numerical values.

(a) Linear Regression: $y = mx + b$

- y represents the dependent variable
- x represents the independent variable
- m is the slope of the line (how much y changes for a unit change in x).
- b is the intercept (The value of y when $x=0$)



$\rightarrow$ Linear regression is a Supervised learning algorithm used to predict continuous value by fitting a straight line to data

b) Polynomial regression : $y = b_0 + b_1x + b_2x^2 + b_3x^3 + \dots$

Polynomial regression is used when the data is not straight line (non-linear).

→ curved line [✓]

c) Naive Bayes :

Naive Bayes is a probabilistic classification algorithm based on Bayes' theorem that assumes features are independent.

d) Gradient Boosting :

Gradient boosting is a model that builds step by step, where each new model fixes the errors of the previous one.

e) Lasso regression :

Lasso regression reduce overfitting and can remove less imp features

f) Ridge regression :

Ridge regression reduce overfitting by making coeff smaller

g) Elastic regression :

Elastic regression is a linear model, where lasso and ridge combines to reduce overfitting and select imp. features

* overfitting : model memorizes the training data instead of learning patterns.

2) classification

- a) logistic regression:
used to predict categorical outcomes using probability. It predicts yes/no (0 or 1) outcomes
- b) Support vector machine:
It finds the best boundary to separate data into classes.
- c) Decision Tree:-
Splits data into branches to make decisions or prediction
- d) Random forest
that builds multiple decision trees and combines their outputs for better accuracy.
- e) k nearest neighbors:
that predicts a value based on the closest data points.

Unsupervised Learning Algorithm:

Data INPUT → UNLABELED DATA → Hidden struc. → Data Recognition → Data recognition

unsupervised learning is a type of ml in which the model learns patterns and structures from unlabeled data without any predefined output.

• TYPES :-

(i) clustering :-

clustering is an unsupervised learning technique that groups similar data points together based on their feature.

TYPES OF CLUSTERING

* K-means clustering :-

K-means clustering is an unsupervised m.l algorithm that groups similar data points into k different clusters based on their similarity or distance.

k = no. of clusters we want to create.

Working steps :-

1. choose no. of clusters (k)
2. select initial centroids (center point)
3. calculate distance betⁿ data point & centroid
4. Assign points to nearest cluster
5. update centroid position
6. Repeat until clusters become stable.

k=3 Means data will divided into 3 cluster.

* Hierarchical clustering :-

It creates a tree-like structure to show relationship betⁿ data points.

- Agglomerative clustering (Bottom-up)
- Divisive clustering (Top-Down)

* Density Based clustering (DBSCAN)

DBSCAN (Density-Based spatial clustering of Applications with Noise) is an unsupervised clustering alg. that groups data points based on their density.

* Principle Component Analysis (PCA):

PCA is a dimensionality reduction technique that transforms high-dimensional data into fewer dimension while preserving imp info.

It works: create new features called principal components. select components with max variance.

* Linear Discriminant Analysis (LDA)

LDA is dimensionality reduction technique that reduce features while maximizing separation betⁿ diff class.

Ex:- Separating categories like:

- cat vs Dog
- Disease vs disease.

(ii) Association: Association Means finding relationships between items or variable in a dataset.

Rule format: $X \rightarrow Y$

"If X happens $\rightarrow Y$ is likely to happen". {milk $\rightarrow$ etc}

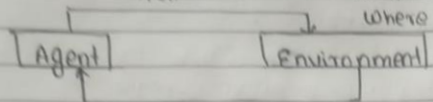
(iii) Dimensionality Reduction:

Dimensionality Reduction Means reducing the no. of features in a dataset while keeping important information.

Why it use:

- $\rightarrow$ Too many features $\rightarrow$ Model slow hunxa
- $\rightarrow$ Overfitting huno sakxa
- $\rightarrow$ visualization difficult hunxa.

3) Reinforcement Learning : Reinforcement Learning (RL) is a type of machine learning where an agent learns by interacting with an environment by taking action & receives reward in return, learning to ~~return~~ improve its action over time.



• Basic concept of reinforcement learning:-

How it works :- A player is playing a game.

1. Agent plays the game (take action)
2. Game give reward or punishment
3. Agent learns which moves are good
4. Improves performance over time.

* Key Terms in Reinforcement Learning

(a) Agent :- The learner or decision-maker that take actions. Example : Robot, AI ~~play~~ player.

→ An agent is the main part of the system that

(b) ~~Environment~~ :- learns and make decisions. It observe environment than choose actions then tries to maximize reward.

(b) Environment :- The environment is everything outside the agent that it interacts with. It gives state information, then return reward after action.
Ex :- Self-driving car → road, traffic, signals

3. State (current situation) :-

A state represent the current condition of the

agent in the environment.

It tells "where the agent is" or "what is happening now"

Ex:

In a game → player position, score

In robot → location, speed, direction.

4) Action:

An action is any move or decision taken by the agent. It depends on current state, leads to new state.

Ex: Move left/right

- Start/stop

→ Turn, Jump

5) Reward (feedback signal)

A reward is the feedback given by the env. after an action. Tells the agent if the action was good or bad.

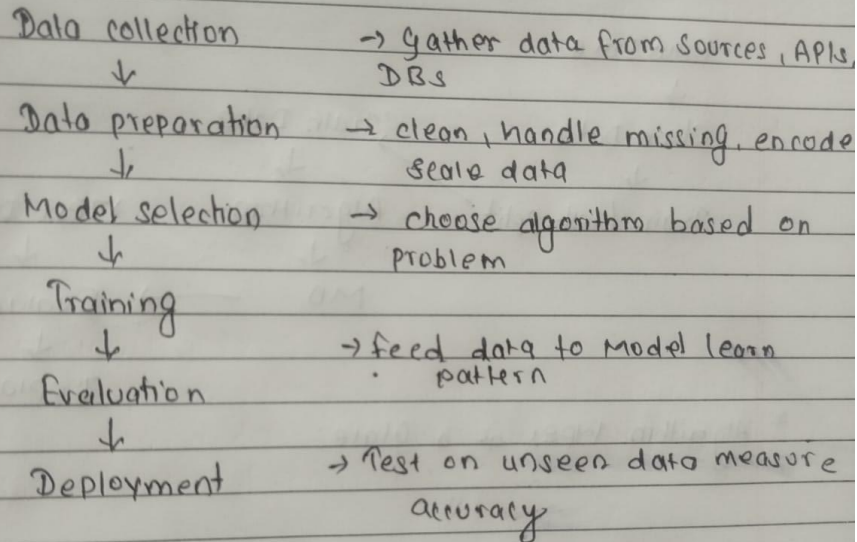
Scikit Learn:

→ A powerful python library for M.L

→ Build on top of Numpy, scipy, and matplotlib.

→ Provides tools for data preprocessing, model building and evaluation.

Machine Workflow - 6 Steps to Success.



Pick the Right Tool:

* Step 1 & 2: Data collection & Preparation:

1) Data collection:

- Gather data from databases, APIs, files, sensors.
- Quality data = Better Model
- More data usually helps.

2) Data Preparation (Most Important!):

- clean: Remove duplicates, fix errors
- Handle Missing: fill gaps or remove rows.
- Enco converter $\rightarrow$ numbers
- Scale: standardize features (mean = 0, std = 1)
- Split: 80% training, 20% test

Garbage In = Garbage out! clean data is essential

* Step 3 & 4: Model selection & Training:

3) Model selection - Pick the Right Tool:

Problem	Algorithm	Why
Predict number	Linear Regression	Simple, interpretable
Yes/No qn	Logistic Regression	Probabilities
Complex pattern	Random forest	High accuracy
Images	Neural Network	Handles pixels
Find groups	k-means	Fast clustering

Not verified

- 4) Training - Model learns:
- Show training data to algorithm. ($x_{\text{train}}, y_{\text{train}}$)
 - Model adjusts internal parameters
 - Minimizes error betn prediction and actual
 - Like student studying from textbook.

• Step 5 & 6: Evaluation & Deployment:

5) Evaluation - Test the model.

Classification:

- Accuracy (% correct)

- Precision & Recall

- F1-score

- Confusion Matrix

Regression

- MAE (avg error) MSE

- (squared error) RMSE

- R^2 (goodness of fit)

(i) Accuracy $\div$ $A_c = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$ ~~$\times 100$~~

Ex: ~~10~~ Out of 10 predictions, the model got 8 correct. So the Accuracy is $(8/10) \times 100 = 80\%$.

(ii) Precision & Recall

Precision = $\frac{TP}{TP + FP}$

FP: exam diya but fail vaye.

Precision is the measure of how many predicted positive cases are actually correct.

Ex: predicted 20 people as sick; out of them:

15 were actually sick (TP), $\text{prec} = \frac{15}{15+5} = 0.75 = 75\%$.

5 were healthy (FP).

• Recall : $\text{Recall} = \frac{TP}{TP + FN}$ i.e. TP = correctly predicted positive
FN = missed positive cases.

iii) F1 Score :

F1 Score is a balance betn ~~predtel~~ precision and Recall. It is used when you want both to be important.

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

iv) Confusion Matrix :

Confusion Matrix is a table used to evaluate a classification model. It shows how many predictions were correct and incorrect.

* Regression : used to predict numerical value.

(i) MAE (avg error) MSE

→ Mean absolute error is the average error.

$$MAE = \frac{|Actual - Predicted|}{n}$$

$$\rightarrow MSE = \frac{(Actual - Predicted)^2}{n}$$

(ii) RMSE (Root mean square Error)

$$RMSE = \sqrt{\frac{(Actual - Predicted)^2}{n}}, \quad RMSE = \sqrt{MSE}$$

(iii) R^2 (R-square) - $R^2 = 1 \rightarrow$ perfect model

Higher R^2 = better model. $R^2 = 1 = \frac{SS_{res}}{SS_{tot}}$

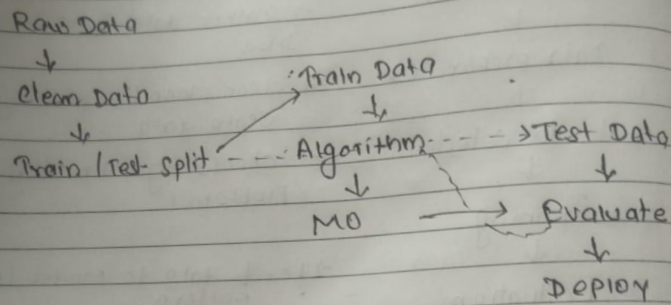
$SS_{res} \rightarrow$ model ko galti

$SS_{tot} \rightarrow$ data ko total variation.

⑥ Deployment - Real world use:

- can Deployment means: trained model for real-world machine use gateway process.
- Convert to API / Web Service:
 - It means making your trained machine learning model accessible so other application can use it.
 - How it works:
 - A User or app Sends a request.
 - The API receive the request
 - The API send data to the model.
 - The model make prediction
 - The API returns the result.
- Monitor performance over time
 - It means continuously checking how well your model is performing after deployment.
- Retrain with new data periodically:
 - It means updating the model by training it again with new data after some time.
 - Why it needed - Real-world data change over time.
- A/B test improvement:
 - A/B testing mean comparing two versions of a model to see which one performs better.

* Complete ML workflow visualization:



* Algorithm types at a glance:

